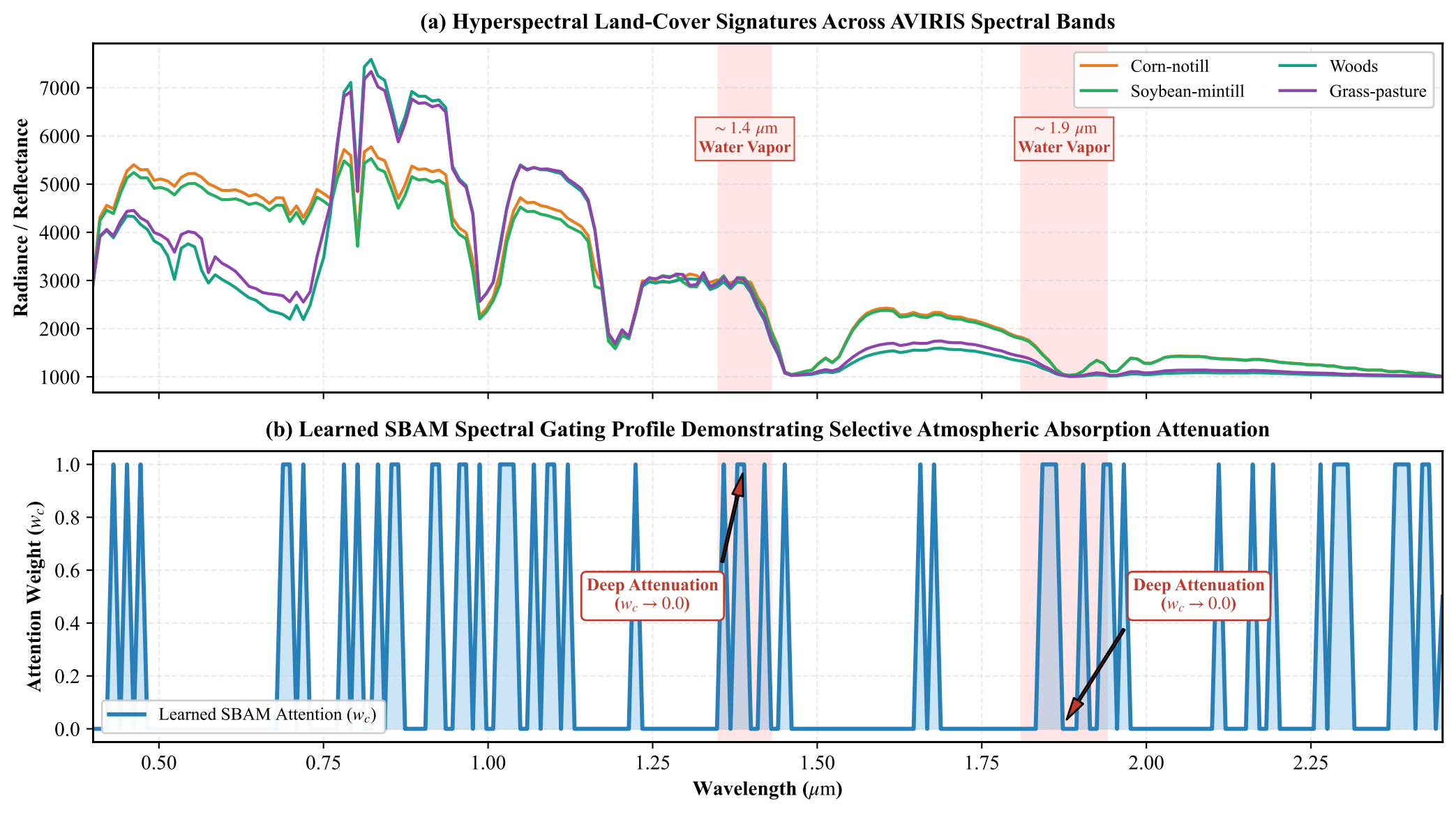

(a) Hyperspectral Land-Cover Signatures Across AVIRIS Spectral Bands
Radiance / Reflectance
7000
6000
5000
4000
3000
2000
1000
Corn-notill
Soybean-mintill
Woods
Grass-pasture
~ 1.4 μm
Water Vapor
~ 1.9 μm
Water Vapor
(b) Learned SBAM Spectral Gating Profile Demonstrating Selective Atmospheric Absorption Attenuation
Attention Weight (w_c)
1.0
0.8
0.6
0.4
0.2
0.0
Learned SBAM Attention (w_c)
Deep Attenuation
($w_c \to 0.0$)
Deep Attenuation
($w_c \to 0.0$)
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
Wavelength (μm)